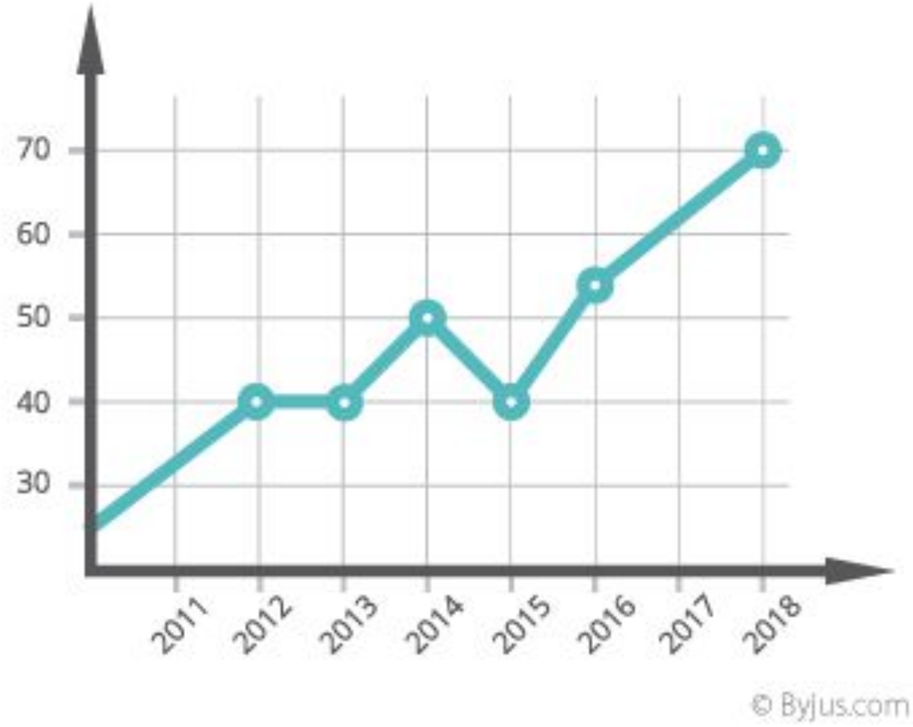
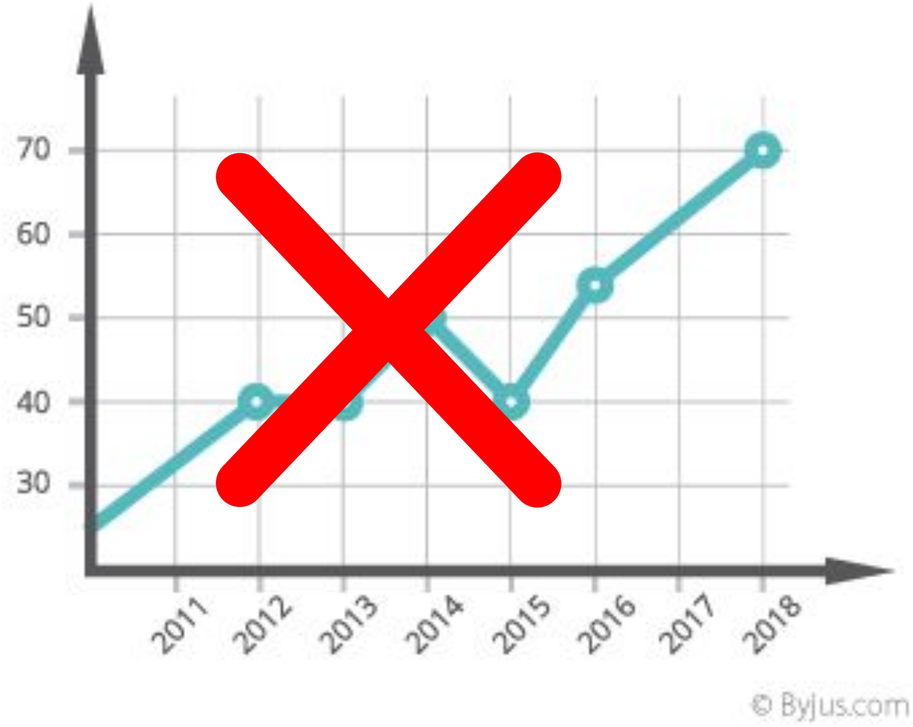
Graphs


70
60
50
40
30
2011
2012
2013
2014
2015
2016
2017
2018
© Byjus.com

Graphs


70
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2011
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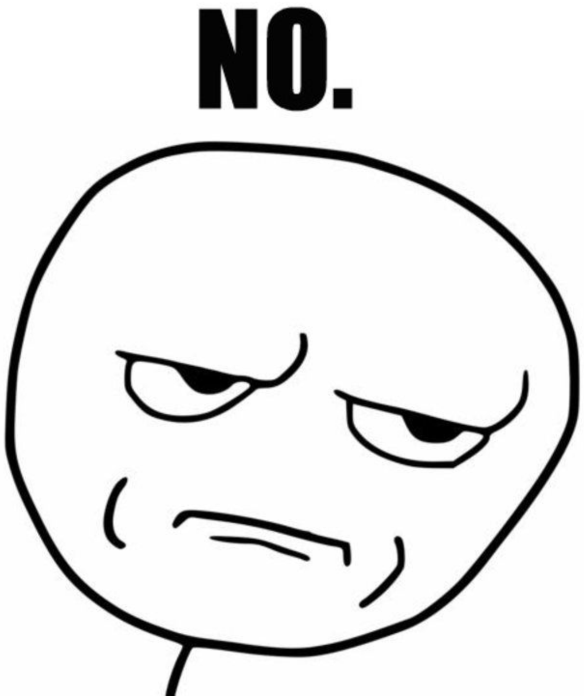
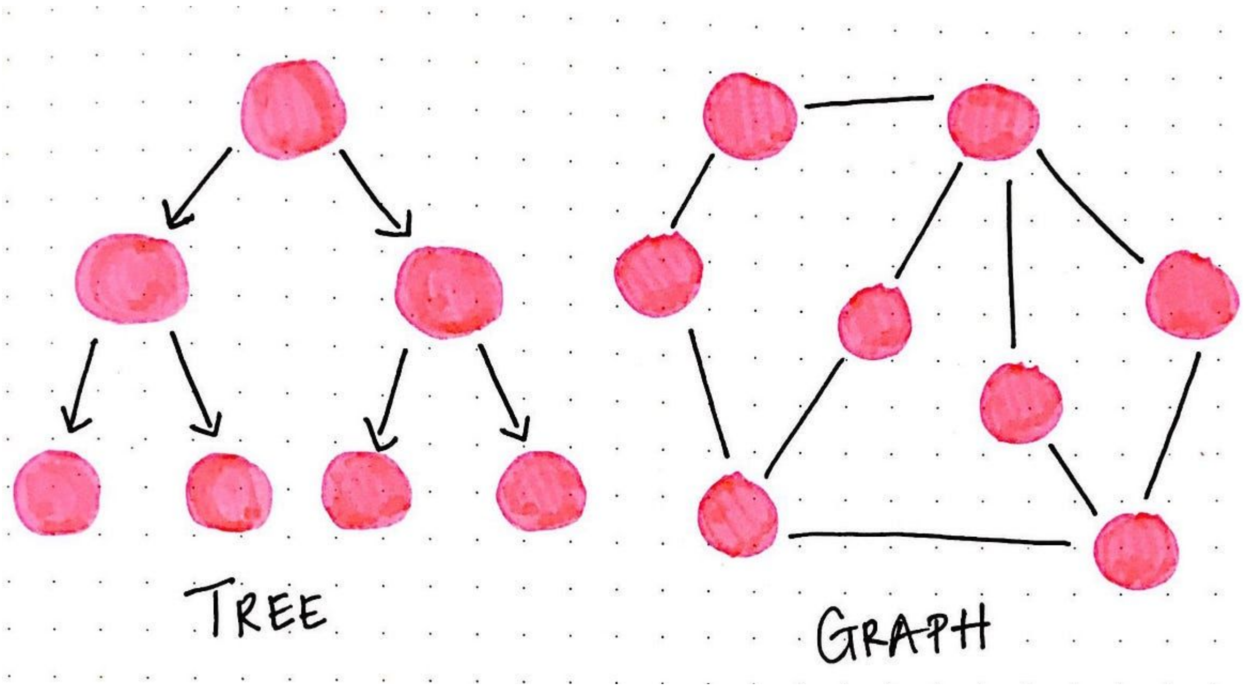
© Byjus.com

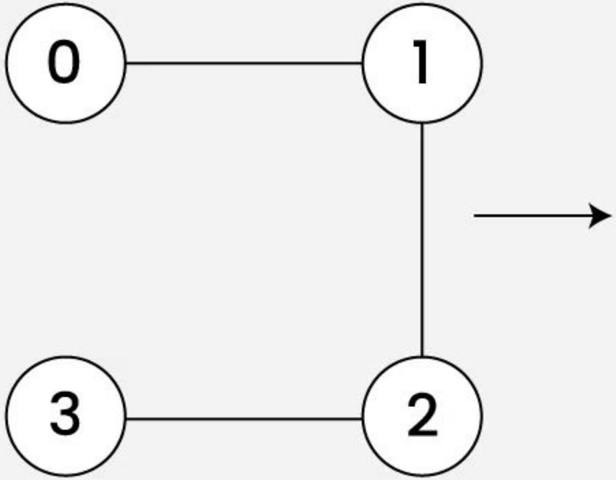

NO.

Memory Representation\\Vl,



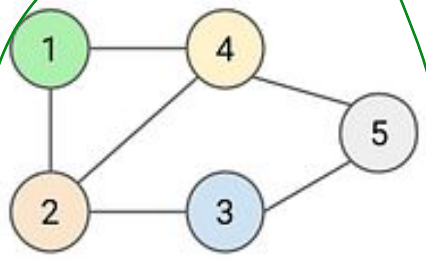
Adjacency Matrix for Undirected and Unweighted graph

Edgelist

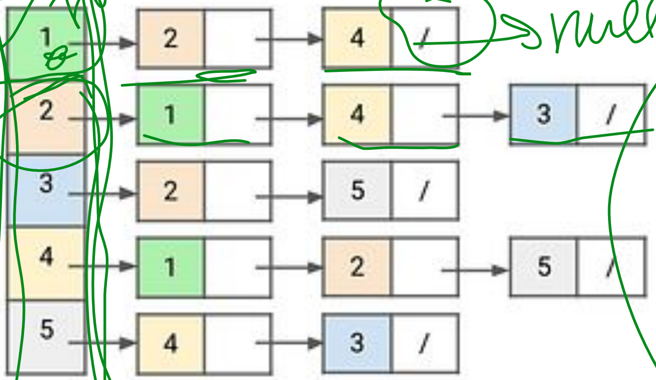


	0	1	2	3
0	0	1	0	0
1	1	0	1	0
2	0	1	0	1
3	0	0	1	0

Adjacency Matrix A[]



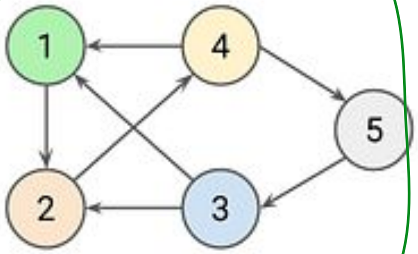
Undirected Graph



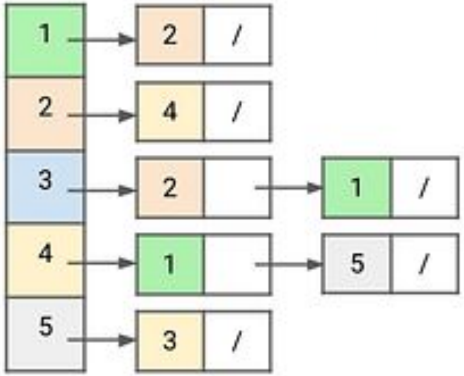
Adjacency List Representation

	1	2	3	4	5
1	0	1	0	1	0
2	1	0	1	1	0
3	0	1	0	0	1
4	1	1	0	0	1
5	0	0	1	1	0

Adjacency Matrix Representation

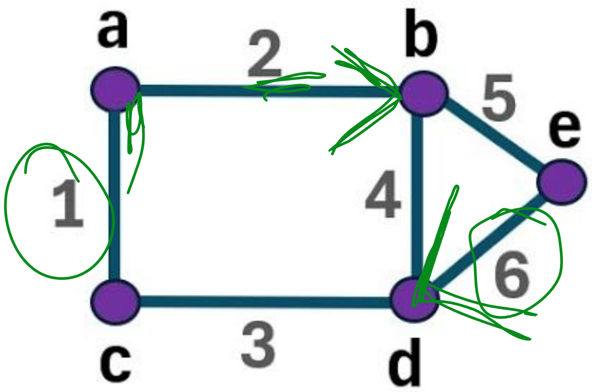


Directed Graph



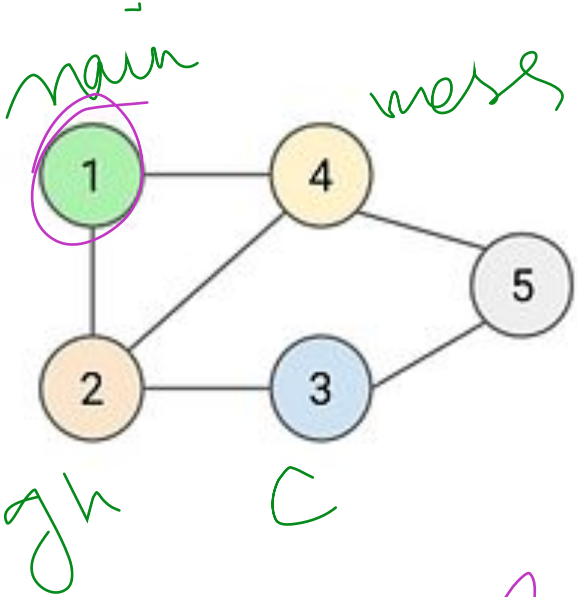
	1	2	3	4	5
1	0	1	0	0	0
2	0	0	0	1	0
3	1	1	0	0	0
4	`1	0	0	0	1
5	0	0	1	0	0

Incidence Matrix



$$
\begin{array}{c}
 \\
\begin{matrix} a \\ b \\ c \\ d \\ e \end{matrix}
\end{array}
\begin{array}{c}
\begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 \end{matrix} \\
\begin{bmatrix}
1 & 1 & 0 & 0 & 0 & 0 \\
0 & 1 & 0 & 1 & 1 & 0 \\
1 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 1 & 1 & 0 & 1 \\
0 & 0 & 0 & 0 & 1 & 1
\end{bmatrix}
\end{array}
$$

BFS



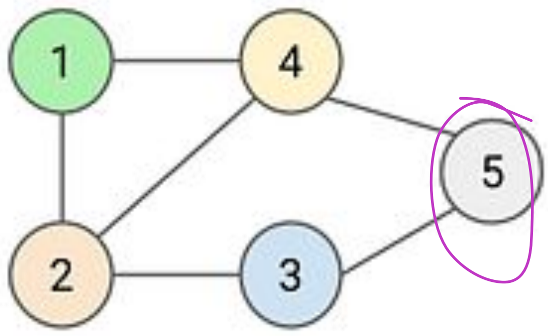
Queue Queue

~~1~~ ~~2~~ 4 1 3

~~1~~ ~~2~~ 4 3

1 2 2 4

DFS



~~5~~

~~2~~ ~~4~~ ~~1~~ 3

54213